

p3_2

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3.2

In [5]:

```
import os, sys, numpy as np, matplotlib.pyplot as plt
sys.path.append('../hw1')
import heat_rod as hr
from mcmc import *
from gibbs_mcmc import *
from scipy.io import loadmat

burnin = 300
samples_gibbs, s2_samples_gibbs, q_map_gibbs, s2_map_gibbs,
posterior_values_gibbs, acc_rate_gibbs =
run_metropolis_gibbs_for_heat_equation(M=10000, burnin=burnin)
```

Shape of observations: (15,)

```
=====
PROBLEM 1.2: METROPOLIS-HASTINGS WITH GIBBS FOR UNKNOWN sigma_squared
=====
```

Hyperparameters: ns = 0.01, sigma_squared_s = 4.0

Initial sigma_squared = 4.0

Running for M = 10000 iterations with 300 burn-in samples

Running Metropolis-Hastings with Gibbs sampling for unknown
sigma_squared...

ns = 0.01, sigma2_s = 4.0

Initial sigma_squared = 4.0

Completed 1000/10000 iterations, MCMC acceptance rate: 0.586,
sigma_squared = 0.2674

Completed 2000/10000 iterations, MCMC acceptance rate: 0.541,
sigma_squared = 0.4642

Completed 3000/10000 iterations, MCMC acceptance rate: 0.538,
sigma_squared = 0.6575

Completed 4000/10000 iterations, MCMC acceptance rate: 0.537,
sigma_squared = 0.4546

Completed 5000/10000 iterations, MCMC acceptance rate: 0.539,
sigma_squared = 0.7179

Completed 6000/10000 iterations, MCMC acceptance rate: 0.536,
sigma_squared = 0.2431

Completed 7000/10000 iterations, MCMC acceptance rate: 0.536,
sigma_squared = 0.5990

Completed 8000/10000 iterations, MCMC acceptance rate: 0.532,
sigma_squared = 0.5049

Completed 9000/10000 iterations, MCMC acceptance rate: 0.529,
sigma_squared = 0.2960

Completed 10000/10000 iterations, MCMC acceptance rate: 0.528,
sigma_squared = 0.2413

Final acceptance rate for q: 0.528

Computing posterior values for MAP estimate...

```
=====
PROBLEM 1.2 RESULTS (with unknown sigma_squared)
=====
```

MAP Estimate (q_MAP, sigma_squared_MAP):

$\Phi = -18.826955 \text{ W/cm}^2$

$h = 0.001960 \text{ W/cm}^2/\text{°C}$

$\text{sigma_squared} = 0.281789 \text{ (°C)}^2$

$\text{sigma} = 0.5308 \text{ °C}$

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```
pi(data|q_MAP,sigma_squared_MAP)pi0(q_MAP,sigma_squared_MAP) =
3.12e-05
```

Posterior Distribution $\pi(q, \sigma^2 | \text{data})$ Summary (after burn-in):

=====

Φ :

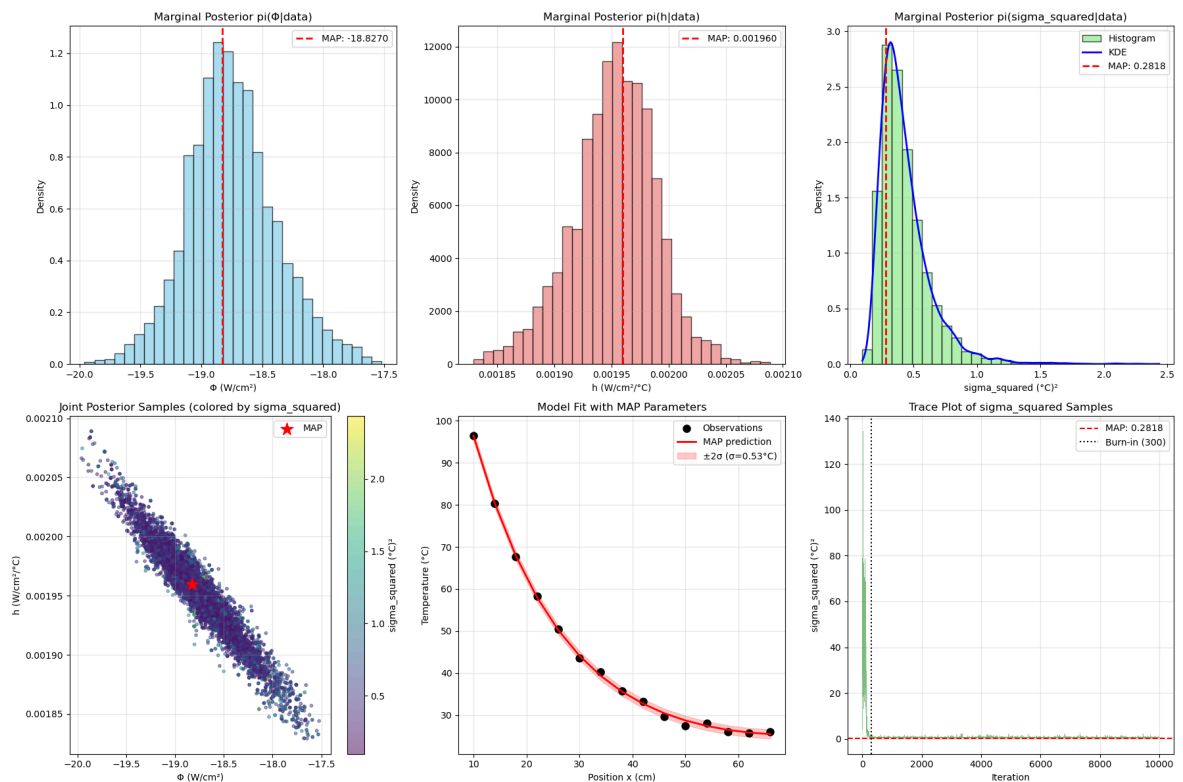
Mean = -18.748429 W/cm²
 Std = 0.363946 W/cm²
 95% CI = [-19.448307, -17.954186]

h:

Mean = 1.952216e-03 W/cm²/°C
 Std = 3.695653e-05 W/cm²/°C
 95% CI = [1.871018e-03, 2.024902e-03]

sigma_squared:

Mean = 0.431687 (°C)²
 Std = 0.205363 (°C)²
 95% CI = [0.192345, 0.951567]

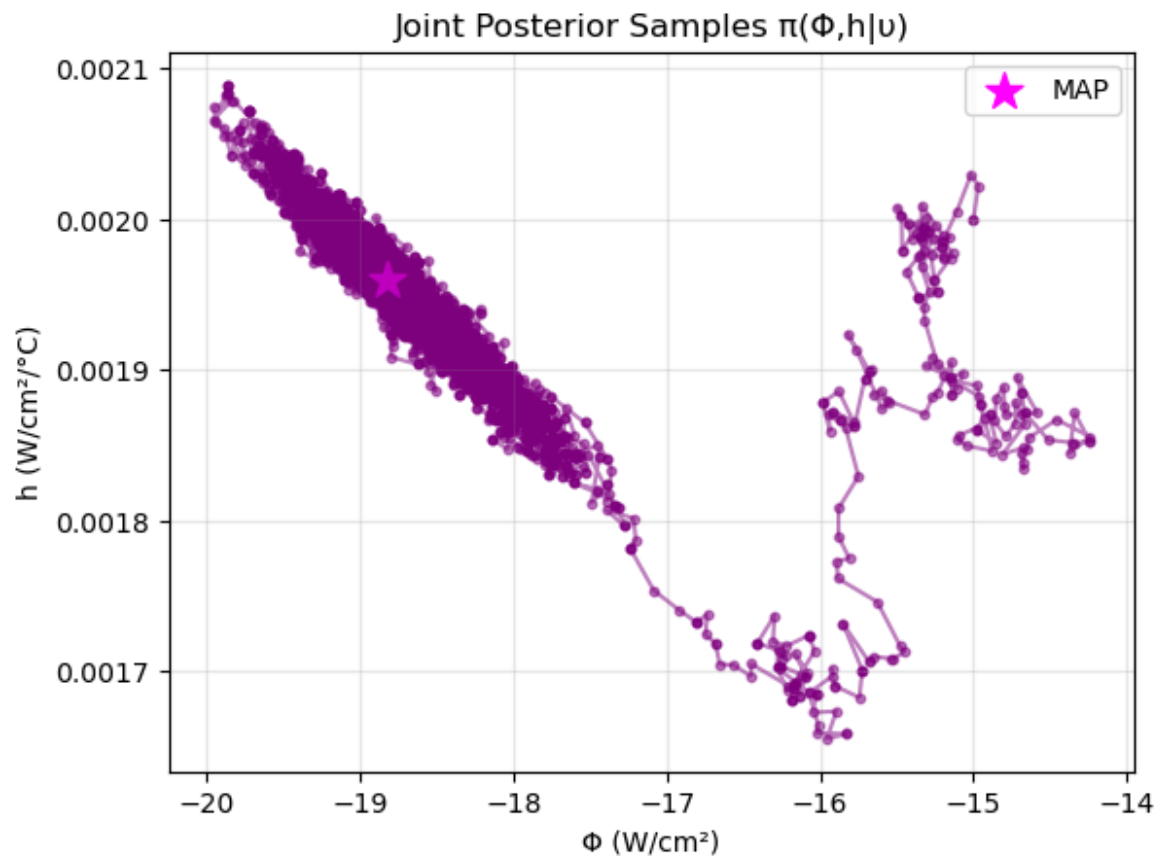


In [6]:

```
plt.scatter(samples_gibbs[:, 0], samples_gibbs[:, 1], alpha=0.5,
            c='purple', s=10)
plt.plot(samples_gibbs[:, 0], samples_gibbs[:, 1], alpha=0.5,
         c='purple')

plt.scatter(q_map_gibbs[0], q_map_gibbs[1], color='magenta', s=200,
           marker='*', label='MAP')

plt.xlabel('Φ (W/cm²)')
plt.ylabel('h (W/cm²/°C)')
plt.title('Joint Posterior Samples  $\pi(\Phi, h|\nu)$ ')
plt.legend()
plt.grid(True, alpha=0.3)
```



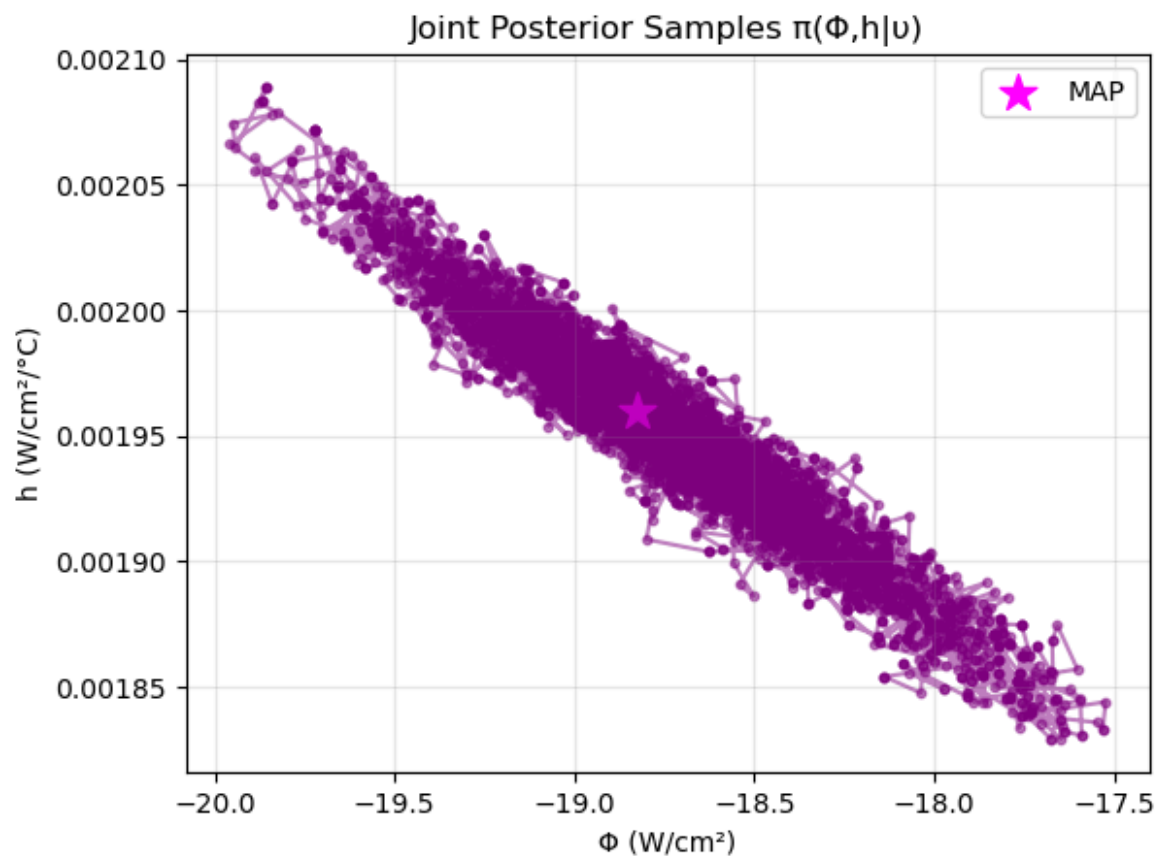
removing burn-in

In [8]:

```
plt.scatter(samples_gibbs[burnin:, 0], samples_gibbs[burnin:, 1],
            alpha=0.5, c='purple', s=10)
plt.plot(samples_gibbs[burnin:, 0], samples_gibbs[burnin:, 1],
         alpha=0.5, c='purple')

plt.scatter(q_map_gibbs[0], q_map_gibbs[1], color='magenta', s=200,
           marker='*', label='MAP')

plt.xlabel('Φ (W/cm²)')
plt.ylabel('h (W/cm²/°C)')
plt.title('Joint Posterior Samples π(Φ,h|υ)')
plt.legend()
plt.grid(True, alpha=0.3)
```



${}^{22}q_{MAP, Gibbs} = [-18.83 W / cm^2; h = 0.00196 W / cm^2 / ^\circ C]$ with ${}^2\sigma^2 = 0.282(^{\circ}C)^2$ after removing 300 burn-in samples.

In [10]:

```

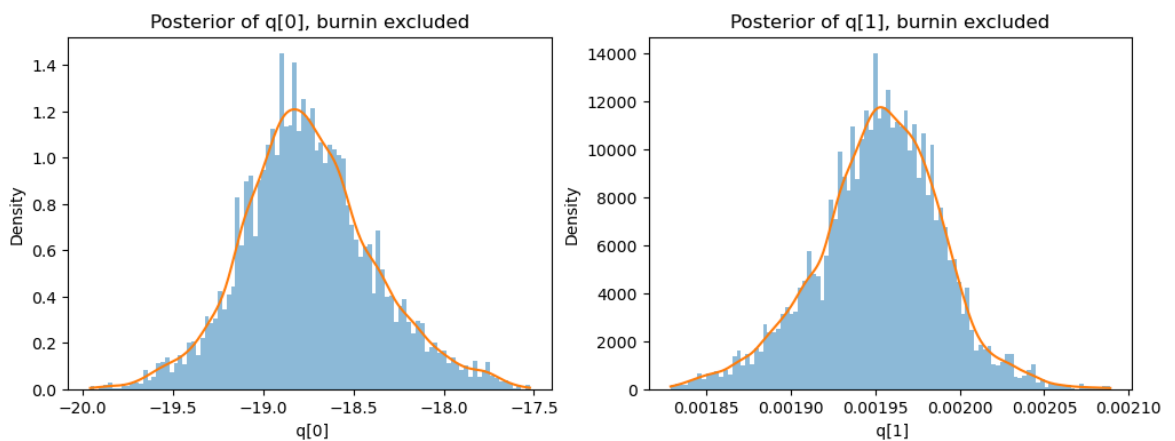
from scipy.stats import gaussian_kde
fig, axes = plt.subplots(1, 2, figsize=(12, 4))
for i in range(2):
    s = samples_gibbs[burnin:, i] #####TODO: BURNIN INCLUDED?

    # Histogram
    axes[i].hist(s, bins=100, density=True, alpha=0.5)

    # KDE smooth curve
    kde = gaussian_kde(s)
    x = np.linspace(min(s), max(s), 300)
    axes[i].plot(x, kde(x))

    axes[i].set_title(f"Posterior of q[{i}], burnin excluded")
    axes[i].set_xlabel(f" q[{i}]")
    axes[i].set_ylabel("Density")

```



In [11]:

```

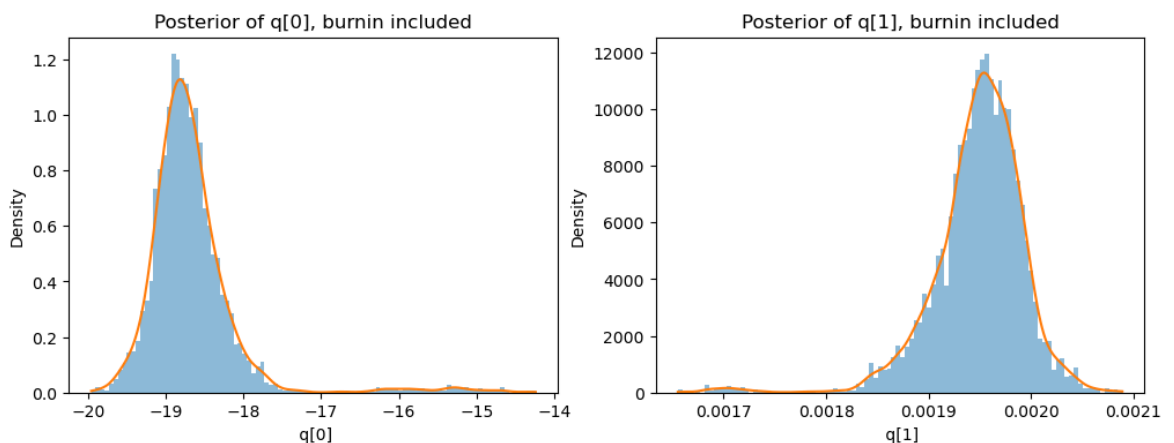
from scipy.stats import gaussian_kde
fig, axes = plt.subplots(1, 2, figsize=(12, 4))
for i in range(2):
    s = samples_gibbs[:, i] #####TODO: BURNIN INCLUDED?

    # Histogram
    axes[i].hist(s, bins=100, density=True, alpha=0.5)

    # KDE smooth curve
    kde = gaussian_kde(s)
    x = np.linspace(min(s), max(s), 300)
    axes[i].plot(x, kde(x))

    axes[i].set_title(f"Posterior of q[{i}], burnin included")
    axes[i].set_xlabel(f" q[{i}]")
    axes[i].set_ylabel("Density")

```



Note: $q[0]$ is ϕ , while $q[1]$ is h .

The exclusion of burn-in samples leads to more symmetric distributions because the early iterations of the MCMC chain are heavily influenced by the initial starting point and have not yet converged to the true posterior distribution. During burn-in, the chain is still "warming up" and exploring regions far from the high-probability area, often exhibiting transient behavior, high autocorrelation, and directional drift as it moves from the initial guess toward the stationary distribution. These early samples tend to be biased toward one side of the posterior (depending on the starting location) and can skew the histogram, creating asymmetry and heavier tails. Once these non-representative samples are removed, the remaining samples are drawn from a more stationary distribution where the chain has reached equilibrium and is properly sampling around the true posterior mode. This equilibrium sampling produces a more symmetric distribution that better reflects the actual uncertainty in the parameters, as the chain has now explored the parameter space in a balanced manner without the directional bias inherent in the transient phase.

3.3

The posterior distributions for both Φ and h exhibit smoother characteristics with reduced variance when σ^2 is estimated rather than fixed. This smoothing effect occurs because the Gibbs sampler allows the model to adaptively adjust the error variance based on the current parameter estimates, effectively weighting observations according to their consistency with the model. Mathematically, this adaptivity arises from the conditional structure:

$$p(\mathbf{q} \mid \sigma^2) \propto \exp\left\{-\frac{1}{2\sigma^2} \sum_{i=1}^n (\epsilon_i - f_i(\mathbf{q}))^2\right\}$$

, $\pi_0(\mathbf{q})$

$p(\sigma^2 \mid \epsilon, \mathbf{q})$

$\ln \Gamma\left(\frac{n_s + n}{2}\right) - \frac{n_s \sigma_s^2 + SS_{\mathbf{q}}}{2}$

The burn-in period becomes more critical in the Gibbs implementation, typically requiring more iterations to achieve stationarity as the chain must now explore the joint space of both parameters and the error variance simultaneously. The Markov chain in the Φ - h plane shows increased exploration during early iterations as the uncertainty in σ^2 propagates through the acceptance probability calculations:

$$r(\mathbf{q}^k, \mathbf{q}_{k-1}) = \frac{\pi(\epsilon \mid \mathbf{q}^k) \pi(\sigma^2_{k-1}) \pi_0(\mathbf{q}^*)}{\pi(\epsilon \mid \mathbf{q}_{k-1}) \pi(\sigma^2_{k-1}) \pi_0(\mathbf{q}_{k-1})}$$

The most significant improvement appears in the model's ability to quantify uncertainty—the credible intervals for predictions properly expand when the model fits poorly and contract when the fit is good, whereas the fixed σ^2 approach maintains constant uncertainty bands regardless of local fit quality.

The Gibbs step adds minimal computational overhead per iteration—only requiring calculation of SS_q and one gamma random variate. However, the improved mixing and more realistic uncertainty quantification often justify this small additional cost. The marginal posterior of σ^2 itself provides valuable diagnostic information: if the true error variance differs substantially from the assumed fixed value, the σ^2 samples will concentrate away from that value, revealing model misspecification. Additionally the Gibbs samples are always accepted (no computational resource waste due to rejected samples).

In practice, estimating σ^2 leads to more robust inference when the true error variance is unknown, which is generally the case in real applications. The fixed σ^2 approach risks either over-confident predictions (if the assumed variance is too small) or unnecessarily conservative estimates (if assumed too large). The Gibbs sampling approach automatically calibrates this trade-off based on the data.